

Manasa Vijayaraghavan

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SUMMARY

M.S. Applied Data Science at USC · Built agentic AI for BFSI at Wipro and production ML at Ford Motor Company and iNextLabs · Hands-on in LLMs, RAG, Agentic AI, multimodal ML, and MLOps at scale · Patent filed · Published author · Open to Fall 2026 internships and full-time from 2027 across the USA.

EDUCATION

University of Southern California M.S. Applied Data Science	Los Angeles, CA May 2027
SRM Institute of Science and Technology B.Tech. Computer Science Engineering	Chennai, India Jun 2025

TECHNICAL SKILLS

Languages	Python, SQL, R, JavaScript, Bash, Git/GitHub
AI & GenAI	LLMs, Agentic AI, RAG, Prompt Engineering, Fine-tuning (LoRA/QLoRA), Multimodal AI, NLP, Computer Vision
ML Frameworks	PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace, LangChain, LangGraph, OpenCV, FastAPI
Data Engineering	PySpark, PostgreSQL, FAISS, pgvector, ETL Pipelines, Feature Engineering, EDA
MLOps & Cloud	MLflow, Docker, CI/CD, GCP (Vertex AI), AWS, Azure AI, Automated Retraining
Certifications	AWS Data Analytics · IBM Data Scientist · Microsoft Azure ML · Google Prompt Design (Vertex AI)

EXPERIENCE

Wipro — AI Native Intern	June 2026 – August 2026
- Built AgenticOS BFSI Skills, an agentic decision-support library for Banking, Financial Services & Insurance, owning two life-insurance skills end to end. - Architected multi-step LangGraph agent workflows with a human-in-the-loop gate enforcing maker-checker approval on every run. - Engineered deterministic Python scoring engines for underwriting tiers and fraud detection, confining the LLM to extraction and narration to eliminate hallucination on key figures. - Implemented real document ingestion (PDF/XLSX) and wired skills to data stores over read-only MCP servers. - Validated with golden-case regression suites and an examiner-traceable audit trail on every run.	
Ford Motor Company — Data Science Intern	Feb 2025 – May 2025
- Built a multimodal emotion analysis system (Python, DeepFace/OpenCV, VADER, Vosk) fusing video, audio, and text into a unified classifier deployed on GCP Vertex AI — improving system efficiency by 75%. - Collaborated with a cross-functional corporate AI team applying NLP, computer vision, and speech processing to real-world multimodal data.	
iNextLabs — Data Science Intern	Aug 2024 – Oct 2024
- Built NLP, anomaly detection, and GenAI solutions (PyTorch, Azure AI) integrated into production, reducing issue resolution time by 40%. - Designed and deployed AI analytics pipelines and dashboards serving 500+ users across technical and non-technical teams.	

PROJECTS

SignMate AI — Patent Filed / <i>Python, PyTorch, OpenCV, CNNs</i>
Real-time Indian Sign Language → text translation; patent filed with IP India and published.
Finance Analytics & NL→SQL Intelligence / <i>PostgreSQL, pgvector, LangChain, FastAPI</i>
ETL and forecasting with plain-English SQL over a pgvector backend, letting non-technical users query financial data.
Large-Scale Customer Churn Prediction / <i>PySpark, XGBoost, SHAP, K-Means</i>
92.6% AUC on 1M+ rows with behavioral segmentation and SHAP explainability on a Plotly dashboard.

PUBLICATIONS & WRITING

AIVerify: An Enhancement of AI Security	Medium
LLM safety, red teaming, and responsible AI — IMDA Singapore Project Moonshot.	
Building a Predictive Model for Cancer Detection	Medium
Random Forest vs. Linear Regression for high-stakes healthcare ML classification.	